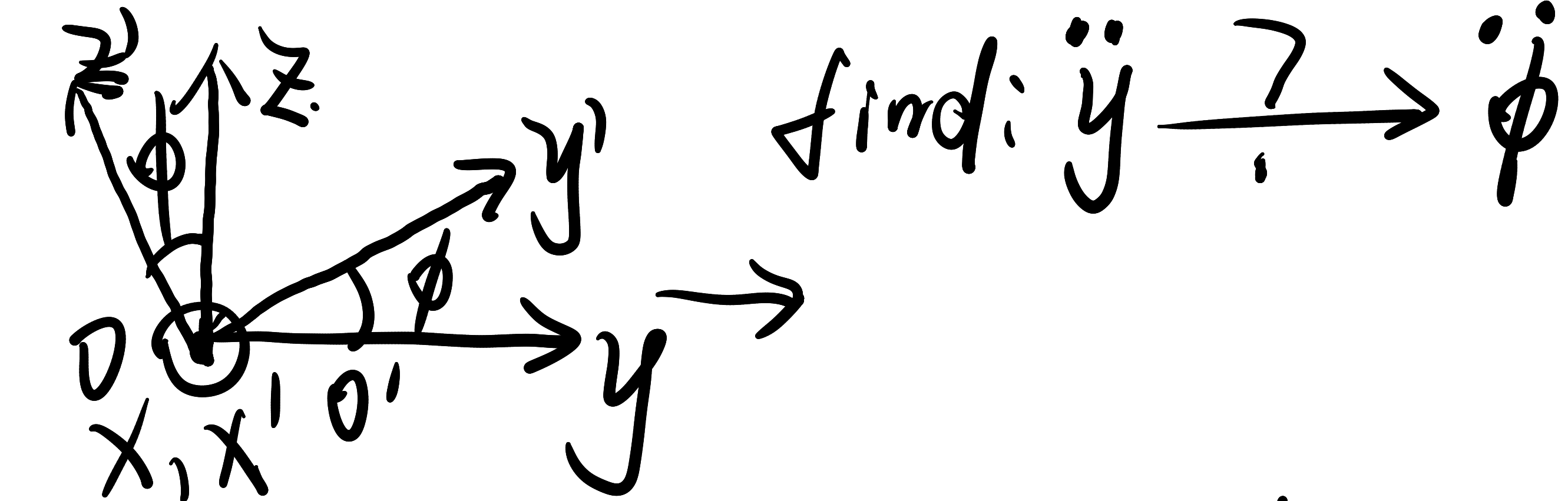




y : fixed, with ground

$x'y'z'$: rotational with car body

$$\vec{a}_{\text{rel}} = \vec{a}_0 + \vec{\omega} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) + \vec{a}_r + 2\vec{\omega} \times \vec{v}_r$$

in the situation:

$$\vec{a}_0 = \ddot{y} \vec{a}_y, \quad \vec{\omega} = \dot{\phi} \vec{a}_x, \quad \vec{\dot{\omega}} = \ddot{\phi} \vec{a}_x, \quad \vec{a}_x = \vec{a}_x$$

$$\vec{r} = L_p \cdot \vec{a}_z, \quad \vec{a}_r = 0, \quad \vec{v}_r = 0$$

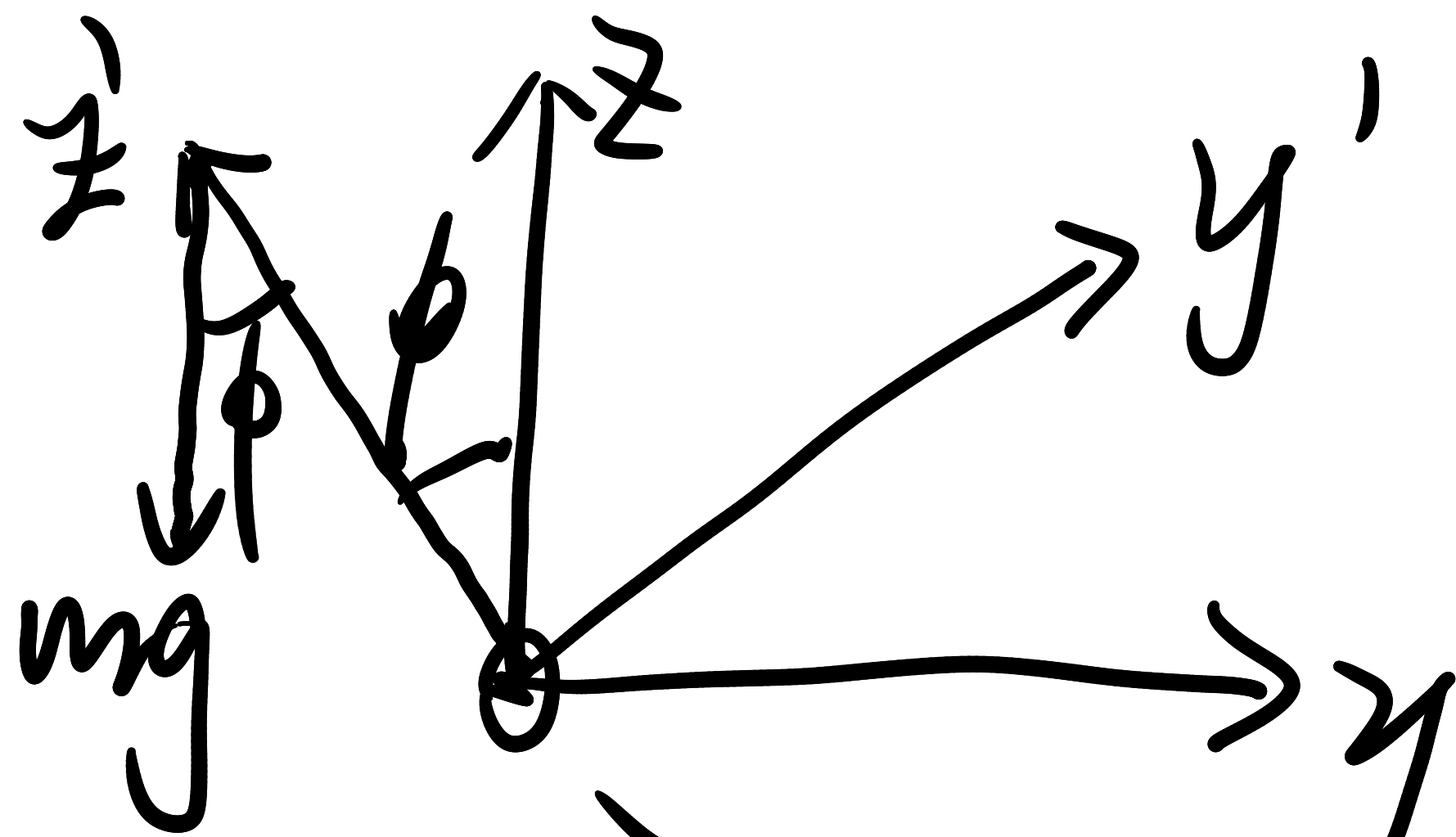
$$\vec{\omega} \times \vec{r} = \dot{\phi} L_p (\vec{a}_x \times \vec{a}_z) = -\dot{\phi} L_p \vec{a}_y$$

$$\vec{\dot{\omega}} \times \vec{r} = -\ddot{\phi} L_p \vec{a}_y, \quad \vec{\omega} \times (\vec{\omega} \times \vec{r}) = -\dot{\phi}^2 L_p \vec{a}_z$$

$$\therefore \vec{a}_{\text{rel}} = \ddot{y} \vec{a}_y + (-\ddot{\phi} L_p \vec{a}_y) + (-\dot{\phi}^2 L_p \vec{a}_z)$$

$$\vec{F} = m \vec{a}_{\text{rel}}$$

$$\vec{F} = -m g \sin \phi \vec{a}_y - m g \cos \phi \vec{a}_z + F_n \cdot \vec{a}_z$$



$$\vec{F} = -m g \sin \phi \vec{a}_y - m g \cos \phi \vec{a}_z + F_n \cdot \vec{a}_z$$

$$\therefore \ddot{y} \cos \phi - L \ddot{\phi} = -g \sin \phi$$

$$\ddot{\phi} = \frac{\ddot{y} \cos \phi + g \sin \phi}{L}$$